

On measure

October 4, 2026

Abstract

A short account of why sets of measure zero do not matter to an integral, written as a worked example.

1 Introduction

An integral does not see a set of measure zero. Section 2 fixes the notation and Section 3 proves the estimate that makes this precise.

2 Preliminaries

Let (X, Σ, μ) be a measure space, as in [1]. We write $L^1(\mu)$ for the integrable functions.

Lemma 1. *If $\mu(E) = 0$ then $\int_E f d\mu = 0$ for every $f \in L^1(\mu)$.*

By Lemma 1, the set in Figure 1 contributes nothing.

3 The main estimate

Theorem 1. *For $f \in L^1(\mu)$ and $\varepsilon > 0$ there is $\delta > 0$ such that*

$$\mu(E) < \delta \implies \int_E |f| d\mu < \varepsilon.$$

3.1 A covering lemma

Cover the set where $|f|$ is large by countably many intervals, as in Figure 1. Its measure tends to zero as the height grows, by dominated convergence; see [2].

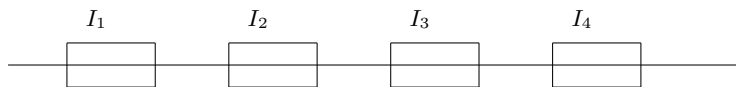


Figure 1: A cover of E by intervals of total length $< \varepsilon$.

3.2 Proof of the estimate

Proof. Write $f_n = \min(|f|, n)$. Choose n with $\int_X (|f| - f_n) d\mu < \varepsilon/2$ and put $\delta = \varepsilon/(2n)$. If $\mu(E) < \delta$ then

$$\int_E |f| d\mu \leq \int_X (|f| - f_n) d\mu + n \mu(E) < \varepsilon.$$

□

4 Two examples

On the line with Lebesgue measure, $f(x) = x^{-1/2}$ on $(0, 1)$ is integrable and unbounded, so δ cannot be taken proportional to ε . For a bounded f it can.

A Notation

$\mu(E)$ is the measure of E ; $L^1(\mu)$ is the space of integrable functions.

References

- [1] Paul R. Halmos. *Measure Theory*. Van Nostrand, 1950.
- [2] H. L. Royden. *Real Analysis*. Macmillan, 3 edition, 1988.